

Understanding the perceived human well-being benefits and values of the Palau National Marine Sanctuary:

A Pilot Study in Airai State



Shirley Koshiba, Supin Wongbusarakum, Lincy Lee Marino, Ikertang Tellei, Andrea Uchel, Randa Jonathan, Louw Claassens, Leah Bukurrou, Dawnette Olsudong, Asap Bukurrou, Matt Gorstein



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Abstract

In 2015 Palau established the Palau National Marine Sanctuary (PNMS), a fully protected offshore large-scale marine protected area (LSMPA) that covers 80% of Palau's EEZ. This LSMPA will provide various direct and indirect benefits to the people of Palau. In an effort to identify and better understand these benefits, a pilot study was initiated to develop a standard approach to: 1) Determine fish resource use patterns and preferences; 2) Identify all human well-being domains and potential benefits from the PNMS as perceived by Palauans; 3) Determine which identified benefits resonates most with the people of Palau and 4) Determine the level of understanding and support of the PNMS by local communities. The most important potential benefits provided by the PNMS were identified during various focus group discussions (FDGs) in 2019. The focus groups included the following key stakeholder groups: male and female fishers, business sector members, PNMS management, and communities. Based on the results from the focus groups, the following well-being domains were identified: sustainable livelihood, food security, economic well-being (income), equitable benefits to all, safety and security of life and property, and culture. A PNMS household questionnaire was developed and structured to focus on: 1) Socio-demographics, 2) Sustainable livelihood and food security, 3) Perceived benefits from PNMS and 4) PNMS knowledge and support. To test the efficacy of the PNMS household survey, a pilot survey of 266 households was conducted in Airai in 2020.

Similar to previous studies, communities were found to prefer reef fish to pelagic fish, with a greater dependence on cheaper and consistently available reef fish. Most respondents agreed with the potential benefits from the PNMS identified during this study, which includes species conservation, replenishment of fish resources, benefits to the nearshore environment and maintenance of Palau's conservation culture locally and internationally. Findings also suggest that even though there is high support for the PNMS, communities have a limited understanding of this LSMPA, which emphasises the need for the development of more effective outreach and education initiatives. Long-term monitoring of potential benefits and perceptions of the PNMS by the people of Palau can now be developed using the standard approach developed here. This will enable tracking of PNMS outreach and education effectiveness, the realisation of potential PNMS benefits to the people of Palau as well as any changes in fish resource use over time.

Introduction

Apart from the well documented ecological benefits from LSMPAs and marine protected areas in general, there are numerous direct and indirect social and economic benefits that can be derived from such conservation efforts. Understanding and quantifying these benefits, especially the less obvious indirect benefits, can, however, be difficult. Owing to the unique characteristics of LSMPAs (including form, function and conceptualisation) a distinct social science research focus is needed (Gruby et al., 2016). In particular, the vastness and in most cases remoteness of LSMPAs results in a misperception that these MPAs do not have a social dimension, as coastal communities are removed from them (Gruby et al., 2016). However, even if mostly devoid of humans, remote and offshore LSMPAs have a human dimension, even if it is different or less visible compared to the human dimension of conventional MPAs, which warrants attention (Gruby et al., 2016).

In 2015, the Palau National Marine Sanctuary was enacted into law by the National government of Palau (Figure 1). This law established an offshore large-scale sanctuary covering 80% of Palau's Exclusive Economic Zone (EEZ). The remaining 20% of Palau's EEZ is zoned as a Domestic Fishing Zone (DFZ), aimed at supporting the development of a sustainable domestic pelagic fishery. After a five-year wind-down period, the PNMS came into full effect in January 2020. The PNMS is one of the largest no-take marine protected areas (MPAs) in the world (475 000 km²), aimed at achieving healthy ocean populations and ecosystems, food security and sustainable development for Palau and its people (PICRC and COS, 2019).

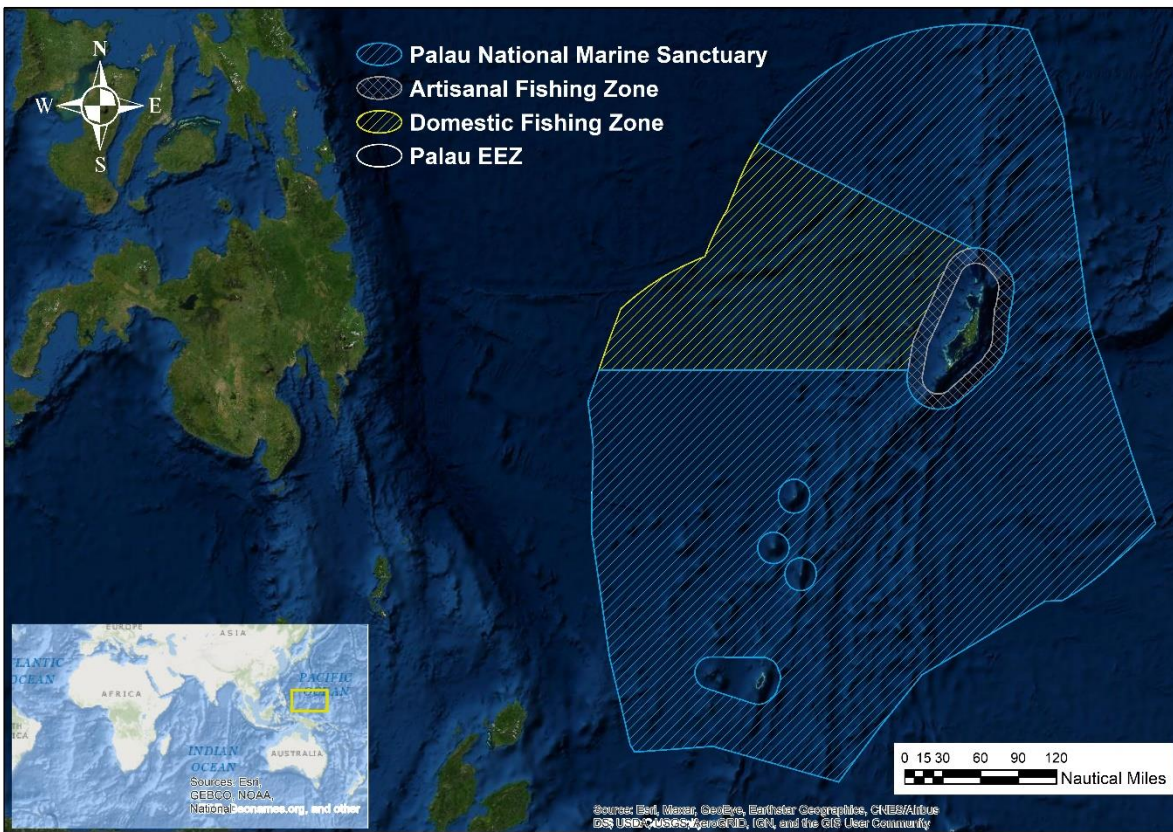


Figure 1. Map of the Palau National Marine Sanctuary which is comprised of the 80% no take Sanctuary, and the remaining 20% Domestic and Artisanal Fishing Zones.

Public support is a critical aspect for successful conservation and illuminating both the direct and indirect benefits that is derived from a conservation action is key to garner that support. Understanding and measuring both the direct and indirect benefits derived from a remote and offshore LSMPA as well as the status of public support for said LSMPA can, however, be quite complex to achieve. The establishment of the PNMS warranted a social research focus to determine and better understand the potential benefits derived from this LSMPA as perceived by the people of Palau. One of the main aims of the PNMS is to ensure food security for the people of Palau with a drive to increase the use of pelagic fish resources. Determining the current fishing behaviour and preferences of communities in Palau and how these might change with the implementation of the PNMS is important in understanding the success of the PNMS in achieving this aim. In addition, an understanding of the level of public support for and understanding of the PNMS is required to determine if ongoing outreach and education initiatives are effective.

To answer and understand the above research questions, a tailored approach was needed to enable standardised monitoring across time and a pilot project was developed to 1) design a tailored household

survey to answer the identified research questions and 2) test the efficacy of the household survey during a pilot survey in the state of Airai.

Methods

Project ethics

This project was conducted with IRB approval from the University of Hawaii (Protocol Number: 2020-00520).

Site description

This study was conducted in the State of Airai, situated on the eastern side of Babeldaob, Palau's main island (Figure 2). Airai is Palau's second most populous state and is made up of 5 main hamlets and a total household population of 868. Airai was identified as a suitable location for the pilot study owing to its proximity to Koror as well as the availability of enough households to survey to ensure statistically robust testing of the approach developed.



Figure 2. Yellow hatched area showing the location of Airai state situated on the southeast of Babeldaob.

Household survey development

Global and Pacific Island literature related to benefits of nature conservation to communities were reviewed (Breslow et al., 2016; McKinnon et al., 2016; Leong et al., 2019). Based on the literature review, an initial list of human well-being domains potentially applicable to the PNMS was developed which included: economic well-being, sustainable livelihood, access to resources, food security, health, education and knowledge, security and safety of life and property, psychological/emotional well-being, subjective quality of life, social well-being, governance, equitable benefits and culture.

To refine this list further and ensure identified domains apply to the Palauan context, a participatory and bottom-up biocultural approach was used (Gavin et al., 2015; McCarter et al., 2018). Focus group discussions (FGDs) were used to understand local stakeholder perspectives in defining human well-being domains and to refine the identified domains to reflect the PNMS. In addition, the FGDs were used to gain an initial understanding of the level of awareness and understanding, involvement and support of the PNMS. A series of focus group discussions was conducted in February 2020 with key stakeholders based in Airai. Stakeholders included: fishers, female and male community members, representatives from the private sector, and stakeholders directly involved with the PNMS. The human well-being domains determined by the stakeholders to be most relevant to the PNMS were: sustainable livelihood, food security, economic well-being (income), equitable benefits to all, safety and security of life and property, and culture.

Based on the eight locally significant domains identified during the focus group discussions, an English and Palauan household survey was developed in July 2020. The draft household survey was pre-tested with 8 households, whereafter it was refined based on feedback received. The PNMS household questionnaire was finalised in August 2020 and included the following four sections: socio-demographics, sustainable livelihood and food security, perceived benefits from the PNMS and PNMS knowledge and support.

Socio-demographic section

This section gathered basic information such as income and education levels and provided insights into the demographics of the sample population.

Sustainable livelihood and food security

Reef fish resources within Palau are under pressure and one of the aims of the establishment of the PNMS and the Domestic Fishing Zone and related domestic pelagic fishing industry was to facilitate and promote a shift from reef fish use to pelagic fish use, to alleviate some of this pressure. The aim of this section was thus aimed at investigating the current status of fishing and fish preferences amongst respondents. Future changes that might take place with regards to fishing and resource use owing to the establishment of the PNMS and Domestic Fishing Zone can then be tracked and compared to this baseline assessment. Questions in this section focused on patterns of fishing (frequency, location, reason) as well as resource use (preferences, reasons for preferences). Questions specifically compared offshore pelagic fishing and fish to reef fishing and fish.

Perceived benefits from the PNMS

This section was aimed at understanding the potential benefits of the PNMS and how these are perceived by the respondents. Potential benefits from the PNMS were identified during the various focus group discussions and presented to respondents during the survey. Respondents were then asked to rate their agreement to and applicability of the different benefits on a scale of 1 (strongly disagree) to 5 (strongly agree). To determine equitability of perceived benefits, survey respondents were asked whether they consider benefits from the PNMS to be distributed equitably amongst all people of Palau, as well as provide suggestions on how benefits can be distributed more equitably. Given the economic stimulus associated with the tourism industry, and the industry's dependence upon healthy marine resources, respondents were also asked how many household members participated in tourism activities over the last 12 months. Activities were defined to include SCUBA diving, snorkelling, water and land tours, and food services (including restaurant) catered to tourists, hotels, etc.

PNMS knowledge and support

This section investigated the level of knowledge and understanding of the PNMS by respondents. To determine the level of awareness of PNMS objectives, respondents were asked to name any objectives of the PNMS to the best of their knowledge and understanding. Respondents were also asked if they had heard, seen and/or participated in any outreach activities related to the PNMS and if so, what platform was used. In addition, the level of support for the PNMS was assessed. The aim of this section was to determine the effectiveness of current PNMS outreach and education initiatives, and to establish

a baseline understanding of the level of knowledge and support for the PNMS, that can be monitored over time to determine the effectiveness of future outreach and education initiatives.

Pilot household survey

To test the efficacy of the PNMS household survey, a pilot survey of 266 households, which constituted 38% of households in Airai, was conducted from October 2020 to January 2021. The latest Republic of Palau census data sourced from the office of Budget and Planning were used for all household information. The survey team, which included staff from the Palau International Coral Reef Center, were trained and served as enumerators. The PNMS household surveys were uploaded to Amazon Kindle Fire 8" tablets and information from respondents were logged electronically. All respondents were 18 years or older and able to participate on behalf of the entire household.

The survey was administered by reading the survey questions to the respondent. Answers from the respondent were then logged into the tablet. Any additional comments or feedback from the respondent were captured by the enumerator and incorporated into the data. Interviews were conducted in either English or Palauan.

Statistical analyses

Survey data were analysed using IBM SPSS Statistics and JASP Statistics Version 0.15.0.0. Frequencies and summary statistics are provided throughout the results section. In addition to providing results of respondent answers to survey questions, an attempt was made to further investigate underlying conditions and drivers for perceptions. This was done by comparing means in t-tests to statistically investigate any significant differences across different respondent characteristics and their perceptions, including perceived PNMS benefits, frequency of seafood consumption, the use of seafood for traditional events, and outreach participation. To determine if different education and outreach approaches have an effect on community perceptions of the PNMS, a Spearman correlation analysis was conducted. Spearman's rho correlation coefficients range from -1 to 1. The closer the value is to 1 or -1, the stronger the correlation. Positive values indicate a positive correlation (e.g., the use of a communication medium is correlated with a more positive perception), and negative values indicate a negative correlation (e.g., the use of a communication medium is correlated with a less positive perception).

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Results

Socio-demographics

The majority (93%) of survey respondents were Palauan citizens, with the remaining 7% comprised of citizens from the following countries: Philippines, USA, Japan, and Bangladesh. Of the respondents who were interviewed (n=263), 53% were female, and 46% were male. The age of respondents ranged from 18 to 92, with an average age of 51 and a mode of 48. Household sizes ranged from 1 to 13 people per household, with an average of 4 people. The majority of household respondents (n=266) reported some level of formal education (Table 1), with 37.2% of respondents reporting at least a college level education.

Table 1: Educational Attainment of Survey Respondents (n = 266)

Level of Education	Number of respondents	Percentage
Elementary	16	6.0%
High school	96	36.1%
Some college (2-year college degree)	99	37.2%
University (4-year, Bachelor's degree)	37	13.9%
Master's degree and/or PhD	16	6.0%
Refused to answer	2	.8%
Total	266	100.0

Almost 40% of respondents (n=266) chose not to provide information related to income and around 32% indicated that they earn between \$5000 and \$20 000 per annum (Figure 3).

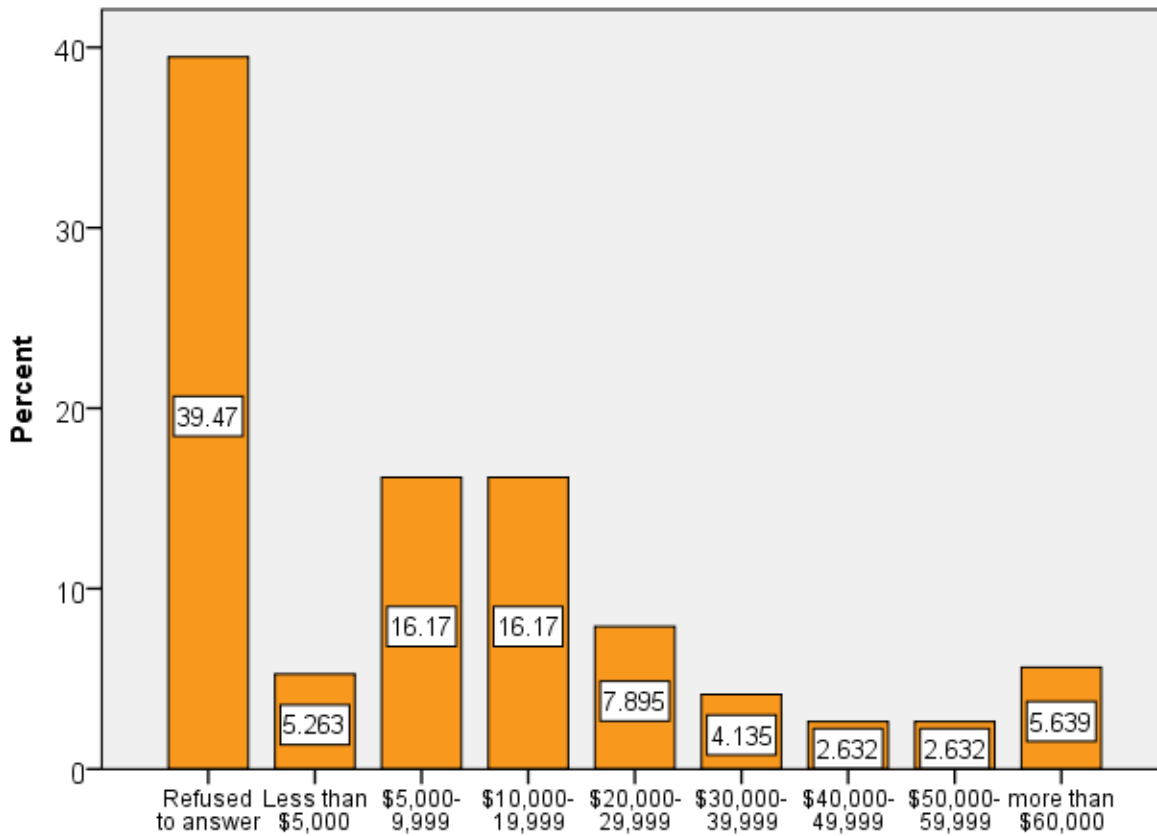


Figure 3. Income profile of respondents (n=266).

Most respondents were either employed by the government (46.2%) or privately employed (20.7%), whilst 30.1% of respondents received their income from either pension or social security payments (Figure 4). Seventy-three percent of respondents listed only one income source, whilst 37% listed multiple income sources.

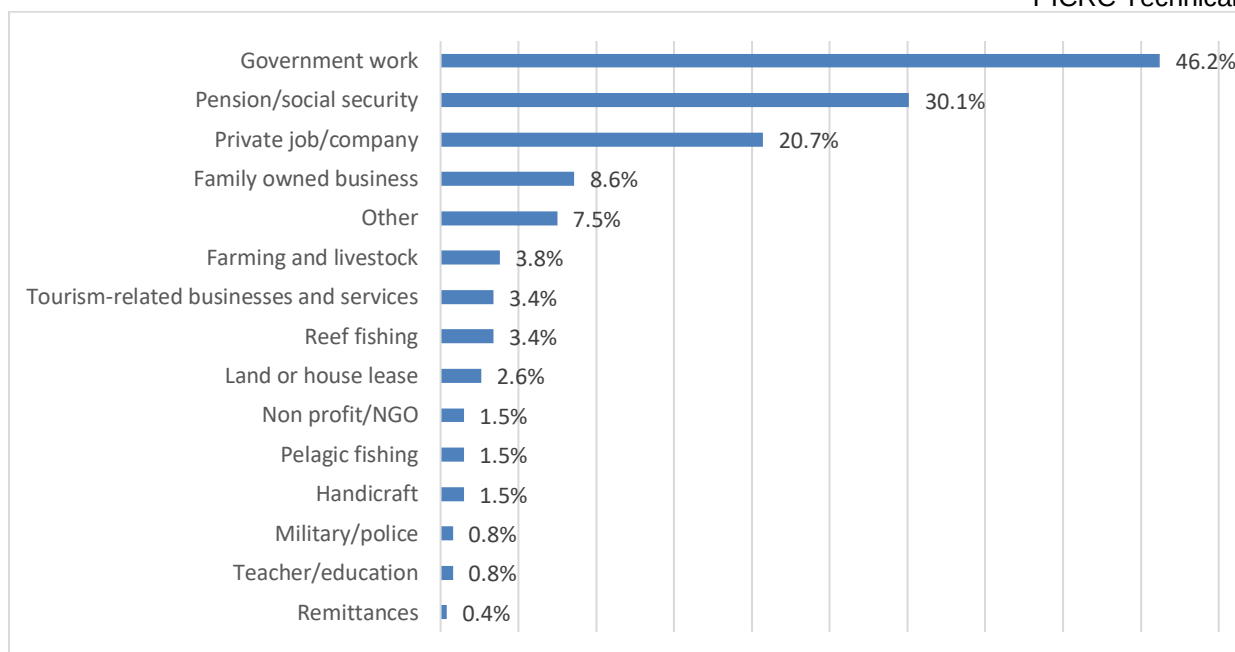


Figure 4: Top Income Sources (n=266)

Sustainable livelihood and food security

Approximately 42% of respondents reported no participation of their households in any type of fishing activities within the past year (Figure 5). Of those respondents that indicated participation in fishing, the majority participated in reef fishing (36.8%), as opposed to pelagic fishing (3.7%) or both pelagic and reef fishing (17.6%) (Figure 5).

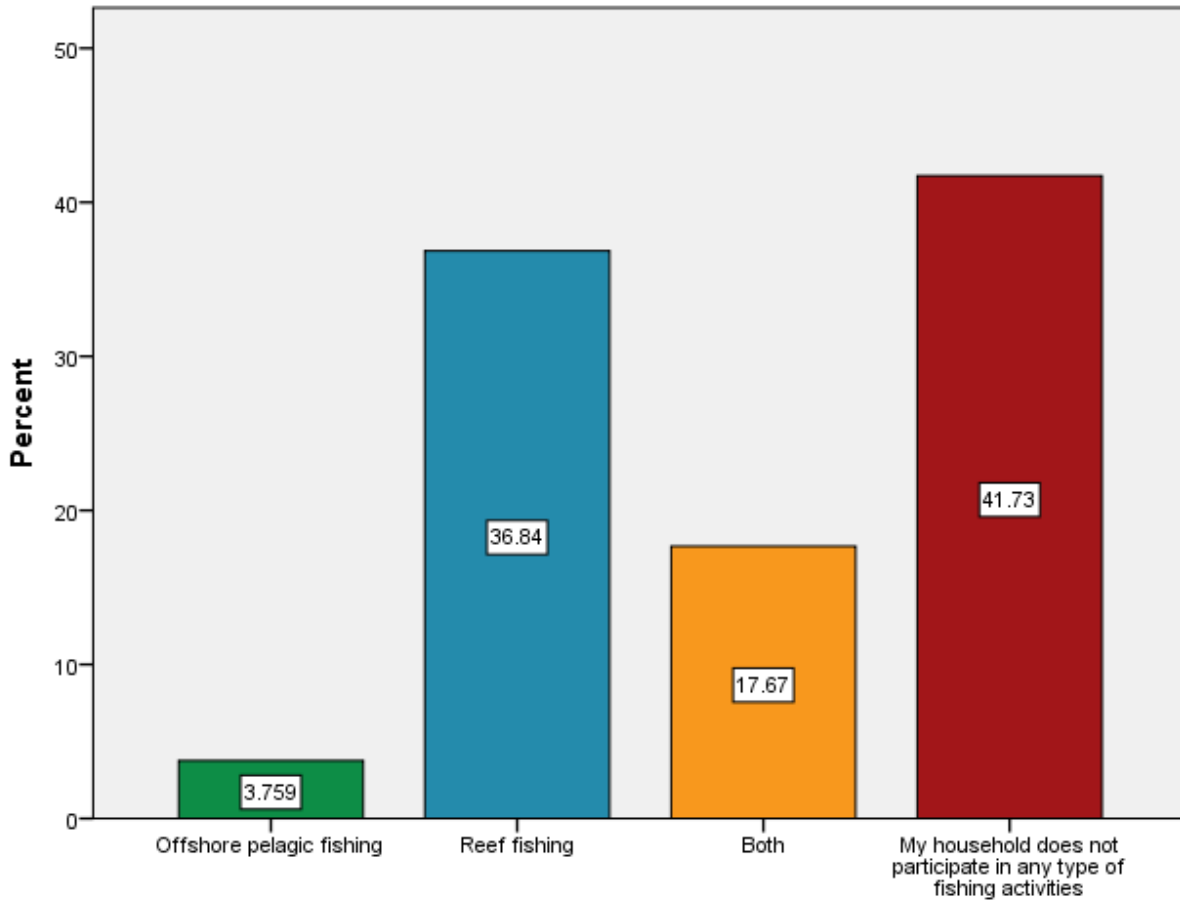


Figure 5. Participation in fishing activity within the past year (n=266).

The frequency of fishing varied between reef and pelagic fishing, with 42.1% of respondents indicating that they rarely fished offshore during the past year, whereas reef fishing took place more often (Figure 6). Between 22-26% of respondents indicated that they fished often both nearshore and offshore (Figure 6).

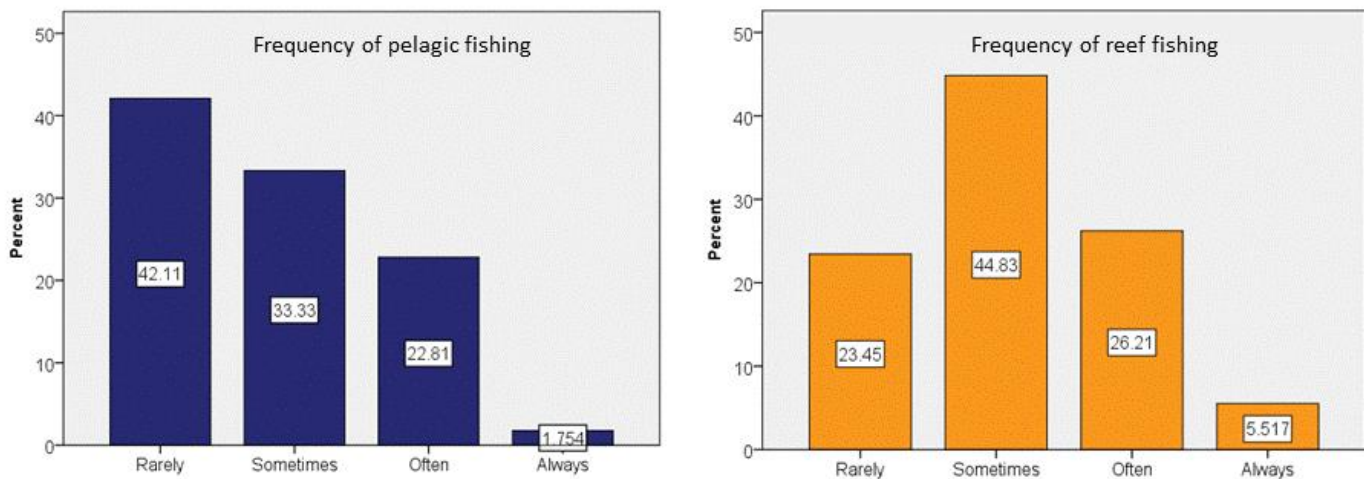


Figure 6. Frequency of pelagic fishing activity (n=57) and reef fishing activity (n=145) within the past year.

Respondents were asked to list the main reasons for fishing and more than one reason could be listed. The major reason for fishing provided by respondents were for household consumption ($\approx 95\%$ for both offshore pelagic and reef fishing) or gifting (86% for both offshore pelagic and reef fishing). Twenty percent of respondents indicated that offshore pelagic fishing is done with the aim to sell the catch, with 17% of respondents participating in reef fishing to sell their catch. Around 50% of respondents listed recreation as a reason for fishing, for both offshore pelagic and reef fishing.

At the household level, more males over 18 years participated in both reef and pelagic fishing, compared to females as well as males under 18 years (Table 2). More females over 18 participated in reef fishing compared to pelagic fishing (Table 2).

Table 2: Number of Males and Females Participating in Fishing-Related Activities at the Household Level (n=266)

	Reef Fishing				Pelagic Fishing			
	Mean	$\pm$ SD	Min	Max	Mean	$\pm$ SD	Min	Max
Males aged 18 and over	0.64	0.87	0	6	0.40	0.72	0	5
Females aged 18 and over	0.31	0.62	0	3	0.17	0.47	0	3
Males under 18	0.14	0.62	0	8	0.07	0.55	0	8
Females under 18	0.01	0.18	0	3	0.03	0.49	0	8

Respondents had a much higher preference for reef fish (54.8%) compared to pelagic fish (15.4%) (Figure 7). Taste was the most important reason for both reef fish and pelagic fish preference (Figure 8). Reef fish was also preferred because of its availability and respondent knowledge of preparation approaches (Figure 8). A subsequent reason for pelagic fish preference was because of its high quality and nutritional value (Figure 8).

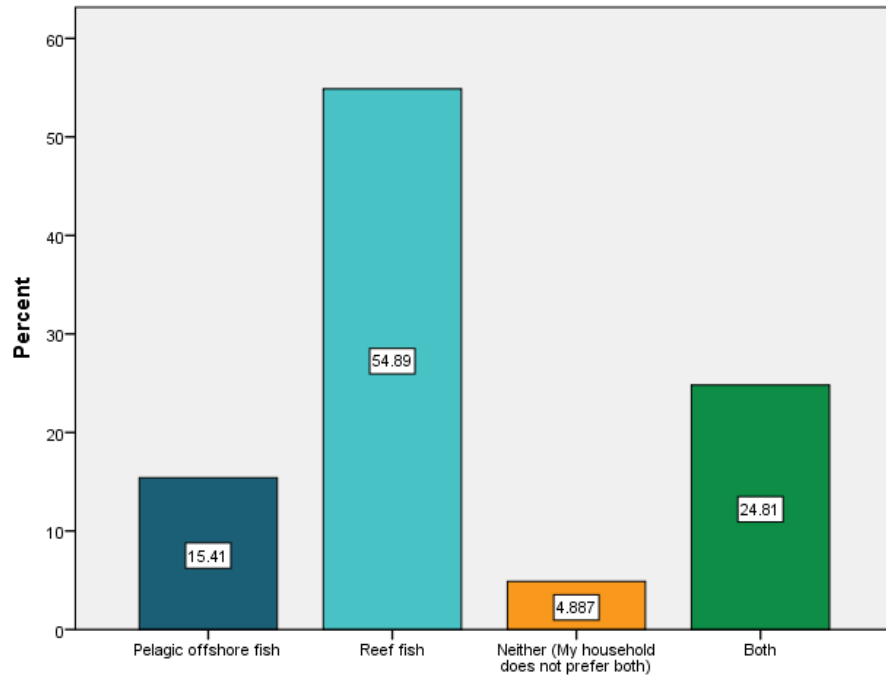


Figure 7. Household preferences for offshore and reef fish (n=266).

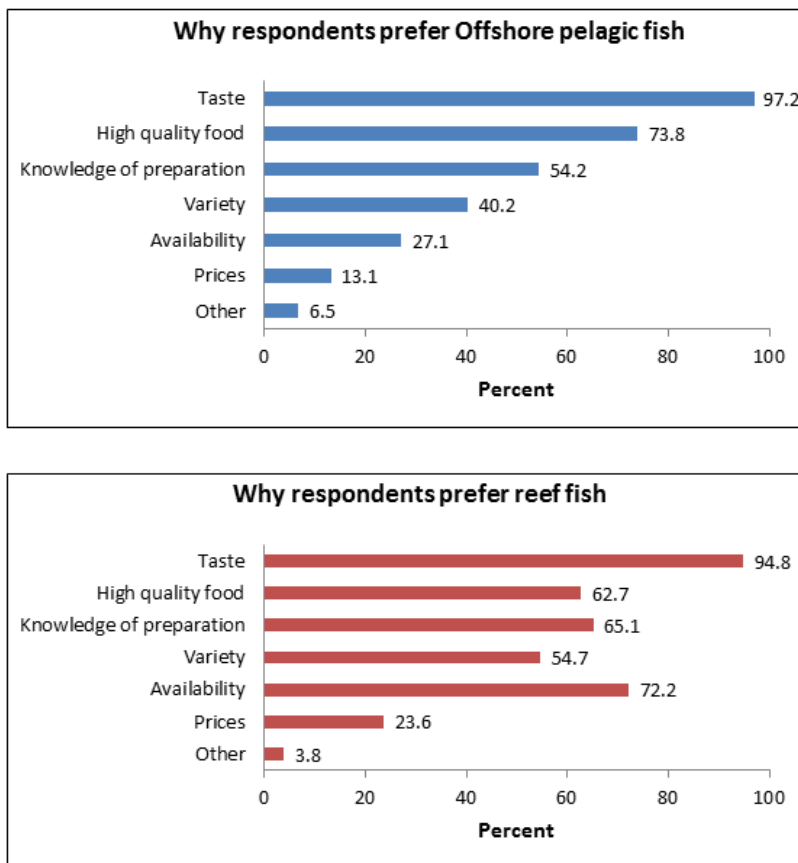


Figure 8. The main reasons why survey respondents prefer offshore pelagic fish (n=107) or reef fish (n=212) for food (more than one reason could be listed by a respondent).

Most respondents (57.7%) that consumed pelagic fish purchase it from markets, although respondents indicated that the supply is not stable and pelagic fish is only sometimes available (Figure 9). Pelagic fish are also commonly given to respondents by friends or relatives (47.4%). Fifteen percent of respondents indicated that they buy pelagic fish from restaurants or through community sellers and 2.4% purchase imported pelagic fish.

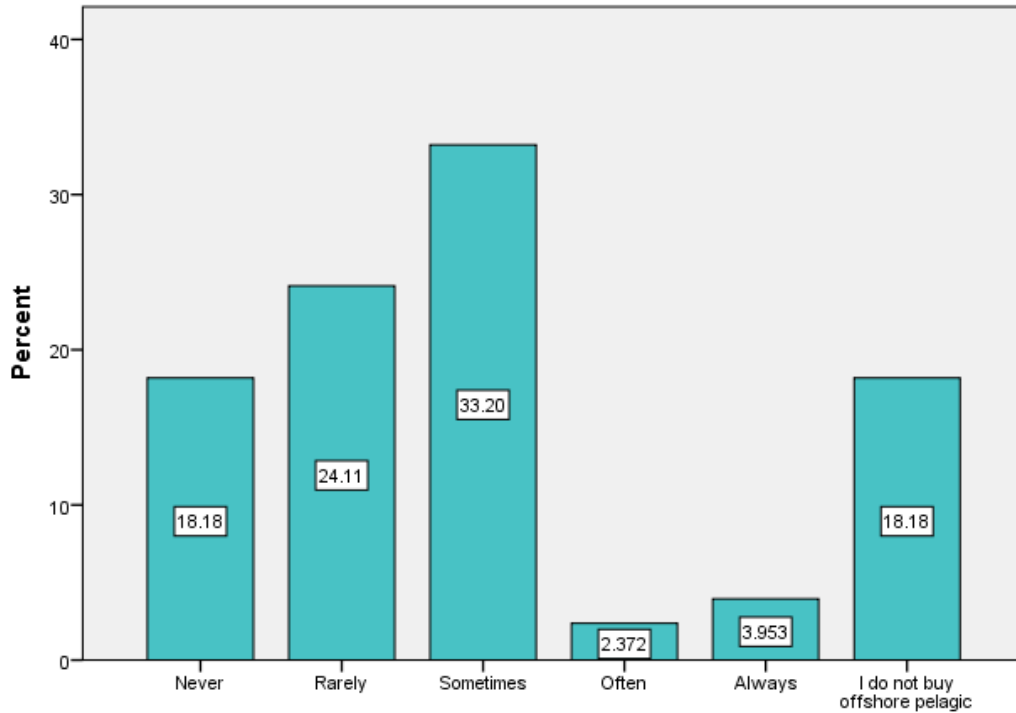


Figure 9. Availability of offshore pelagic fish in the market (n=253).

Over the past year, approximately 34% of respondents consumed fresh reef fish between 3-4 times per week, whereas pelagic fish (mostly tuna) were mostly consumed rarely or never (53.7%) (Figure 10). Other pelagic species (non-tuna) were rarely consumed by most respondents (Figure 11).

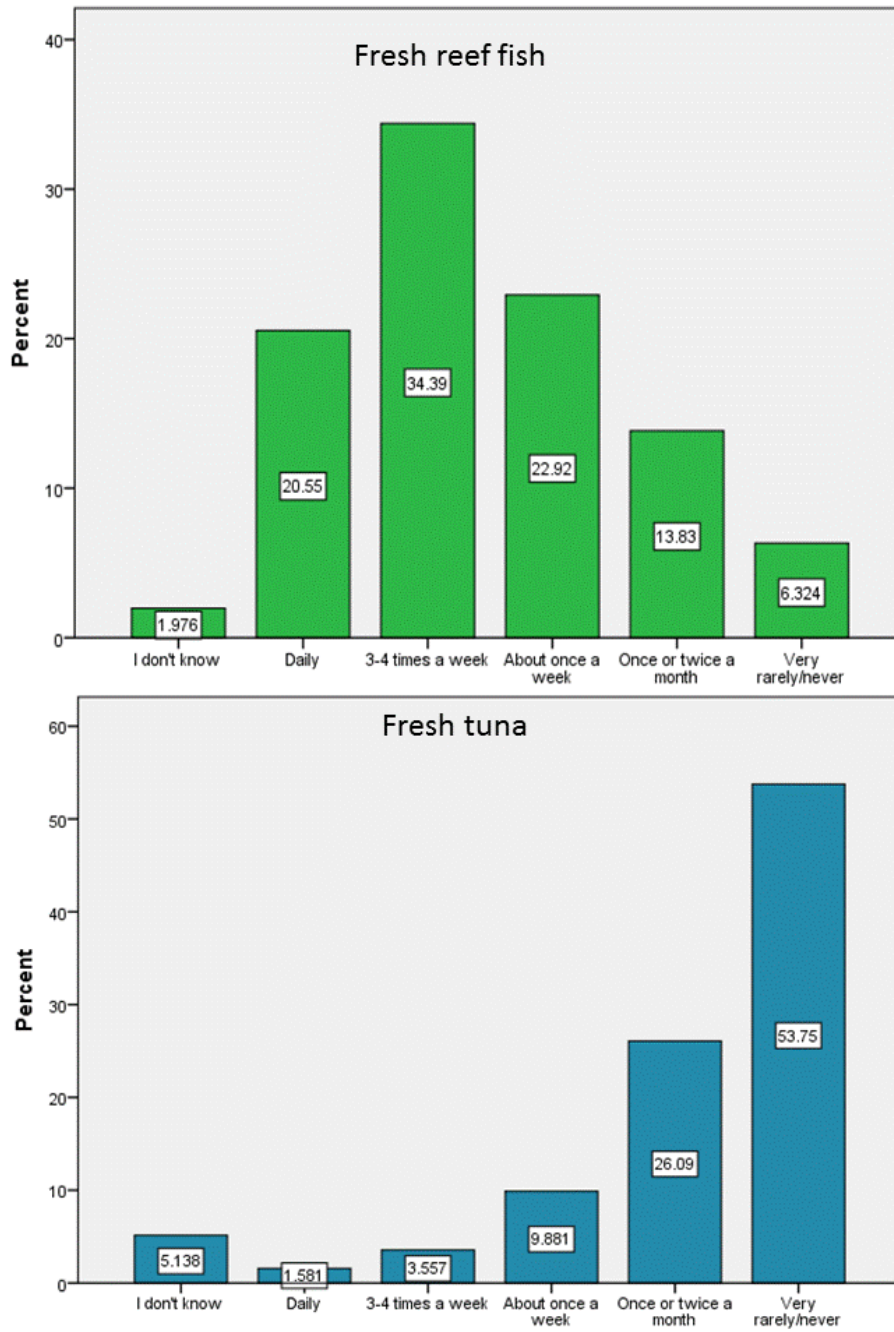


Figure 10. Frequency of consumption of fresh reef fish and fresh tuna within the past year (n=253).

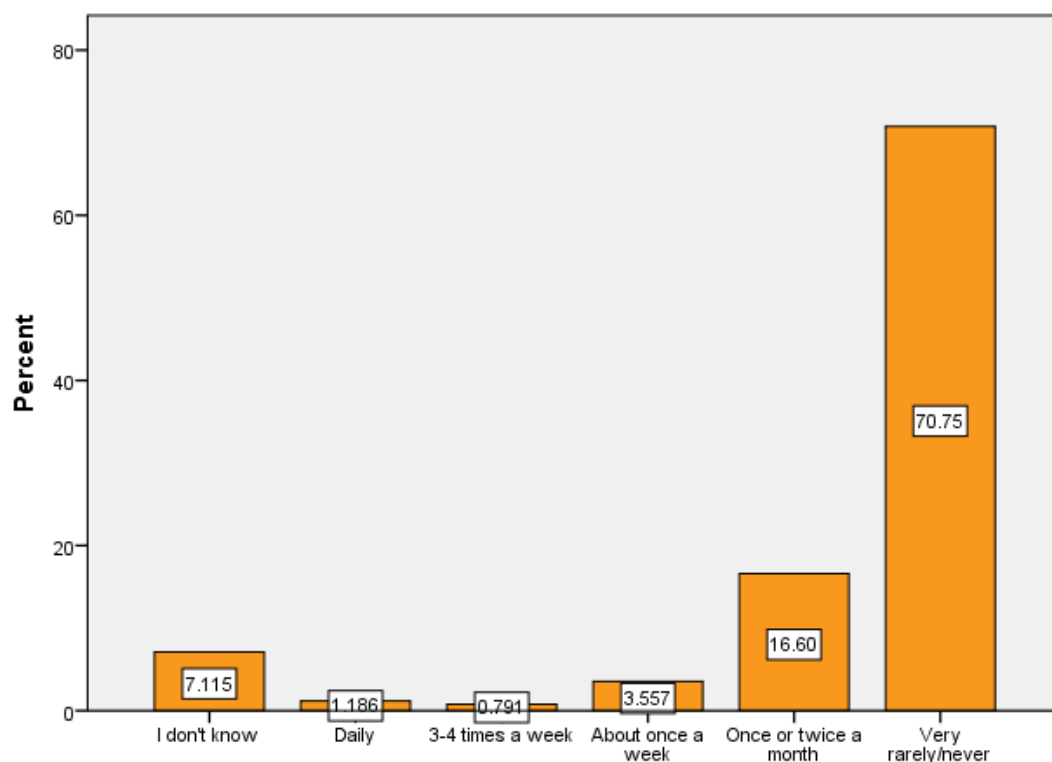


Figure 11. Household consumption of other fresh non-tuna Pelagic fish (i.e. Wahoo, Marlin, and Swordfish etc.) in the past year (n=253).

Other protein sources consumed by respondents included beef, pork and chicken and 32.4% of respondents indicated that their household consumes alternative protein 3-4 times per week. Canned meat, including canned tuna, corned beef etc. were consumed on a daily basis by 26% of respondents (Table 3). Processed reef fish and tuna were not commonly consumed by respondents and only around 13% of respondents indicated consumption once or twice a month (Table 3).

Table 3: Household consumption of various processed foods within the past year (n=253)

Type of food	Daily	3-4 times a week	About once a week	Once or twice a month	Very rarely/never	I don't know
Processed reef fish	4.7	7.1	11.9	13.4	58.5	4.3
Processed tuna	0.8	1.2	4.7	13.4	70.0	9.9
Processed Non-tuna (Wahoo, marlin, swordfish etc.)	0	0.8	2.4	6.7	78.7	11.5
Canned meat (Kansume)	25.7	23.3	19.4	9.9	21.3	0.4
Other meat (beef, pork, chicken)	26.9	32.4	18.2	11.5	9.9	1.2

Reef and pelagic fish were rarely used for cultural purposes by respondents (Figure 12). When used, pelagic fish were used more often compared to reef fish (Figure 12).

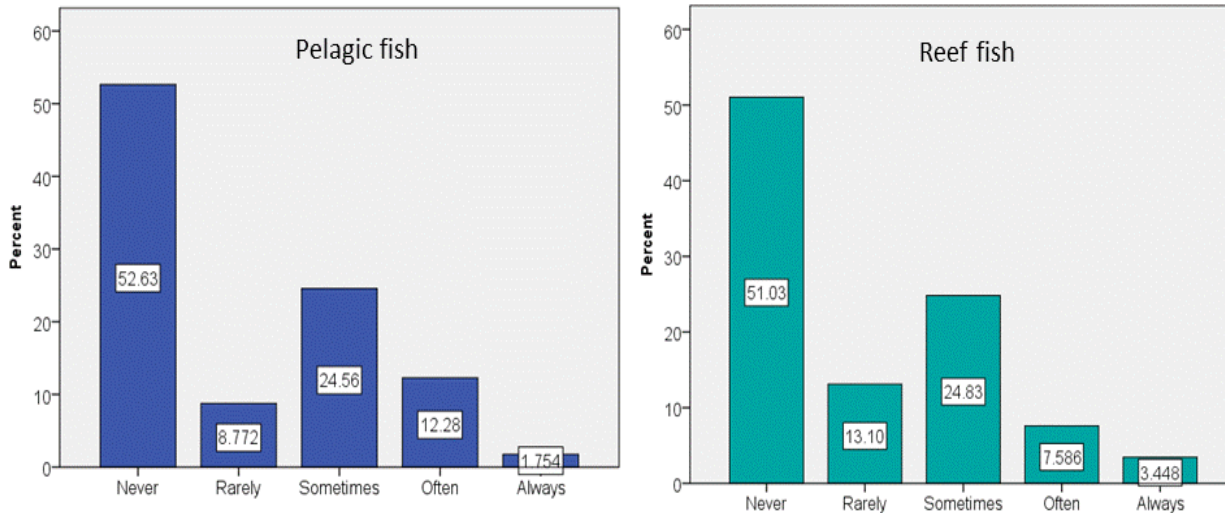


Figure 12. Household use of offshore pelagic fish (n=57) and reef fish (n=145) for customary or traditional events in the past year.

Perceived benefits of the PNMS

The majority of respondents (39.8%) indicated high support for the PNMS and the closure of 80% of Palau's EEZ (Figure 13). About 20% of respondents were, however, uncertain about their support and could not provide an answer (Figure 13). Only about 20% of respondents indicated limited or no support for the PNMS (Figure 13).

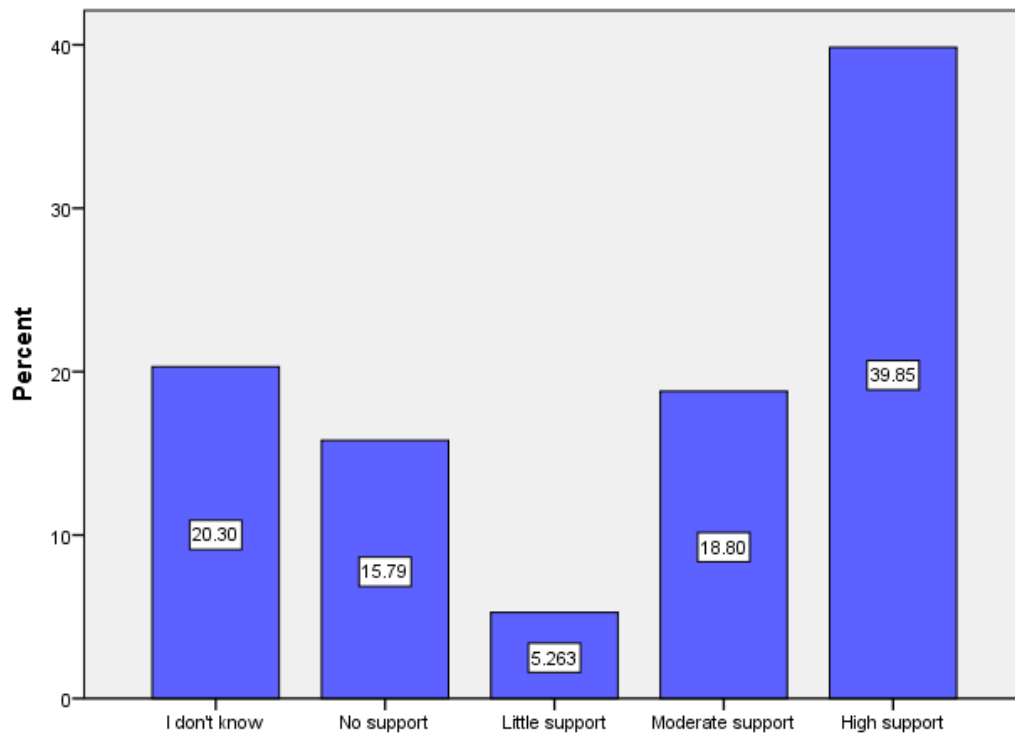


Figure 13. Respondent's level of support for the 80% closure of Palau's Exclusive Economic Zone (EEZ) (n=266).

Table 4 provides a summary of the main reasons for no or little support from the 20% of respondents. Restriction to fishing was listed as one of the main reasons for limited or no support for the PNMS, followed by the idea that the PNMS will not be effective in protecting highly migratory fish species such as tuna.

Table 4. Summary of main reasons participants indicated lack of support for PNMS.

Main reasons stated by participants for indicating “Little” or “No support” for the PNMS
- Palauans should not be restricted from fishing - Migratory fish will not stay in Palau’s waters - PNMS protected area too large/loss of income & revenue for Palauans

Almost half of respondents (49.5%) strongly or slightly agreed that benefits of the PNMS are distributed equally amongst the people of Palau. About 20.7% of respondents stated that they were unable to assess or did not know if benefits of the PNMS are being distributed equally, and 14.7% of respondents indicated they strongly disagree and that benefits are not equally distributed to the Palauan people. About 6.8% of respondents reported they neither agreed nor disagreed with the statement that PNMS has equitable distribution of benefits. Those respondents that indicated that benefits are not equally distributed were asked to identify, to the best of their knowledge, who they think benefits the most and least from the PNMS (Table 5). In addition, respondents were asked to provide suggestions to improve the equitable distribution of benefits from the PNMS (Table 6).

Table 5. Perceived distribution of benefits

Who benefits the most?	Who benefits the least?
Those with access to equipment, boats for offshore fishing	Community/local Palauan people
Political leaders/high officials/government officials	Low income individuals/families
People who work/involved in the PNMS Initiative	Non-fishers/those with no access to fishing equipment/materials

Table 6. Suggestions to improve management of the PNMS

Suggestions to the management of the PNMS for equitable distribution of benefits to people of Palau
Community/outreach & educational outreach programs for schools
Stronger law enforcement, surveillance and monitoring of the PNMS
Funding and/or subsidy program to assist people with limited means/resources to access the PNMS

The mean ($\pm$ sd) level of agreement from respondents to various statements related to the PNMS and its potential benefits are presented in Figure 15. Means provided are on a scale of 1-5. A value of 3 would indicate a relatively neutral position on the corresponding statement. A value greater than 3 would indicate the general sentiment leaning more toward agreement, increasing in strength as the mean approaches 5. Similarly, a value less than 3 would indicate the general sentiment leaning more toward disagreement, increasing in strength as the mean approaches 1 (for detailed results see Table 1A in Annex A).

On average, most respondents strongly agreed or agreed to most of the statements provided, suggesting potential benefits derived from the PNMS as identified during the focus groups discussions were correct (Figure 14). This includes benefits in terms of food security, environmental protection, tourism and traditional connections. Respondents also indicated their support and understanding for the protection of offshore environments and the establishment of the PNMS as a no-take zone to achieve sustainable resource use and environmental conservation. Respondents agreed less with statements regarding PNMS management and the ability of Palau to effectively manage and enforce the PNMS. It is also clear that pelagic fish is not yet provided on a consistent and affordable manner for most households.

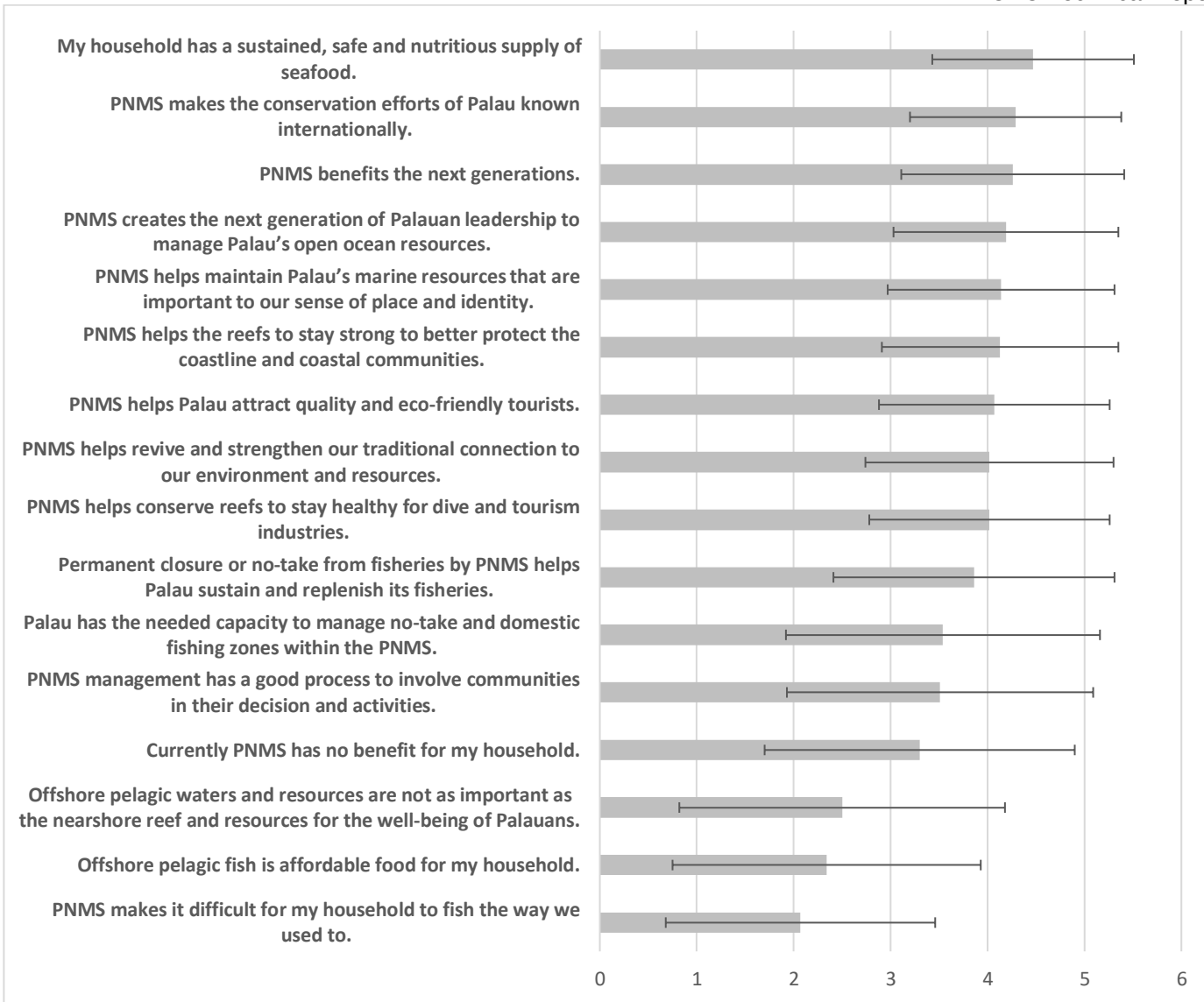


Figure 14. The average ($\pm$ sd) level of support for 16 statements presented to respondents to understand the potential benefits from the PNMS (n=266).

An average of 0.11 adult males, and 0.10 females per household participated in tourism related activities (Table 7). Children also participate, but at lower levels than they do in fishing related activities (Table 2). A maximum of one child participates in tourism related activities at the household level (Table 7).

Table 7. Number of Males and Females Participating in Tourism-Related Activities at the Household Level (n=266)

	Mean	SD	Min	Max
Males age 18 and over	0.11	0.44	0	4
Females age 18 and over	0.10	0.33	0	2
Males under 18	0.01	0.11	0	1
Females under 18	0.01	0.11	0	1

PNMS knowledge and support

Healthy oceans and an increase of marine resources were most listed as objectives for the PNMS by respondents (Figure 15). Maintaining a constant supply of pelagic fish was also listed as an objective of the PNMS by most respondents, although, the development of a domestic pelagic fishery was less common (Figure 15).

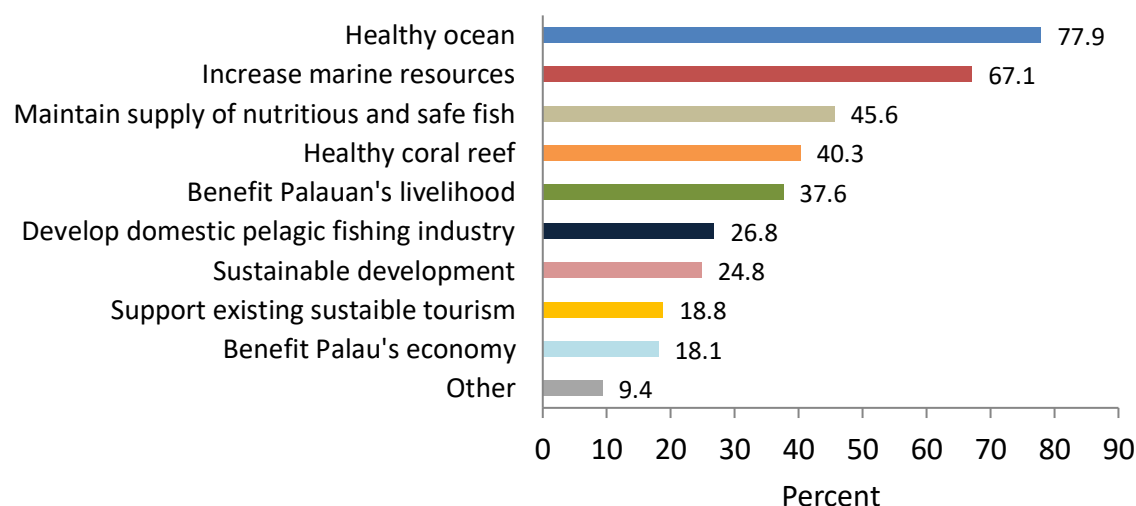


Figure 15. Respondent's level of knowledge of the objectives of the PNMS (n=149).

Television and radio were the two most common platforms with which respondents were informed of the PNMS (Figure 16). Social media platforms, presentations, meetings and newspapers are other approaches whereby respondents received information of the PNMS (Figure 16).

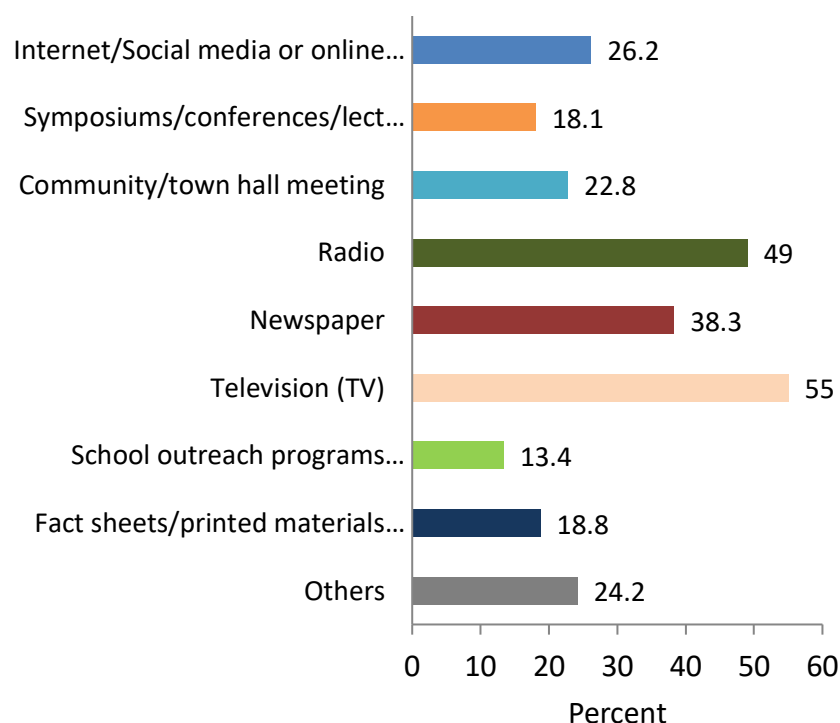


Figure 16. Sources of Information about PNMS (n = 149).

Understanding perceptions of PNMS benefits

When unpacking results further, there are some respondent characteristics that drive certain perceptions of the PNMS. Respondents that use fish for customary or traditional purposes agreed more, on average, that the PNMS benefits the next generation and that it is creating the next generation of Palauan leadership to manage Palau's open ocean resources (Table 8). These respondents also believed that the PNMS provides benefits to their households. This suggests that those who use fish for customary or traditional events may have higher legacy and bequest values as it relates to the PNMS when compared to those who do not use fish for customary or traditional events. Those households that source offshore fish by fishing, showed a greater understanding of the importance of marine protected areas and the PNMS to sustain and replenish fisheries.

Table 8. Perceptions of PNMS for Households That Use Fish for Customary or Traditional Events.

	Uses fish for customary and traditional events		Does not use fish for customary and traditional events		t	p
	n	Mean	n	Mean		
Currently PNMS has no benefit for my household	57	2.842	61	3.557	2.453	0.016
PNMS benefits the next generations	57	4.632	66	4.136	-2.694	0.008

PNMS creates the next generation of Palauan leadership to manage Palau's open ocean resources	56	4.429	66	4.015	-2.101	0.038
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Only statistically significant findings are reported
Answers of "don't know" excluded

Respondents that eat fresh reef fish 3-4 times a week or more, agreed more, on average, that nearshore reefs and resources are more important for the well-being of Palauans (Table 9). Part of this perception could be due in part to affordability concerns related to pelagic fish as these households more likely consider pelagic fish to be unaffordable (Table 9).

Table 9: Frequency of Fresh Reef Fish Consumption and PNMS Perceptions.

	Has eaten fresh reef fish at least 3-4 times a week on average over the last 12 months		Has eaten fresh reef fish once a week or less on average over the last 12 months			
	n	Mean	n	Mean	t	p
Offshore pelagic waters and resources are not as important as the nearshore reef and resources for the well-being of Palauans.	116	2.302	76	2.842	2.208	0.028
Offshore pelagic fish is affordable food for my household.	111	1.937	76	2.842	4.003	< .001

Only statistically significant findings are reported
Answers of "don't know" excluded

Table 10 indicates that those who have heard, seen, read and/or participated in any outreach or awareness activities related to the PNMS tend to agree less with the statements in the survey that are related to perceived PNMS benefits. It should be noted though, that it is not necessarily the case that those who have heard, seen, read and/or participated in any outreach or awareness activities related to the PNMS do not agree that PNMS provides benefits; rather they agree at lower levels when compared to those who have not heard, seen, read and/or participated in any outreach or awareness activities related to the PNMS. This is exemplified by means greater than 3 for the positive sentiment statements. This suggests that the type of communication medium can be important when conducting education and outreach.

Table 10. Participation in Outreach and PNMS perceptions

	Has heard, seen, read and/or participated in any outreach or awareness activities related to the PNMS		Has NOT heard, seen, read and/or participated in any outreach or awareness activities related to the PNMS			
	n	Mean	n	Mean	t	p
Offshore pelagic waters and resources are not as important as the nearshore reef and resources for the well-being of Palauans.	128	2.211	76	2.987	3.270	0.001
Offshore pelagic fish is affordable food for my household.	122	1.959	72	2.986	4.577	< .001
PNMS helps conserve reefs to stay healthy for dive and tourism industries	116	3.784	73	4.397	3.407	< .001
PNMS helps the reefs to stay strong to better protect the coastline and coastal communities.	119	3.891	70	4.543	3.665	< .001
Currently PNMS has no benefit for my household	113	3.540	69	2.913	-2.602	0.010
PNMS helps maintain Palau's marine resources that are important to our sense of place and identity	122	3.951	69	4.464	2.968	0.003
PNMS management has a good process to involve communities in their decision and activities	113	3.301	59	3.915	2.450	0.015
PNMS makes the conservation efforts of Palau known internationally	110	4.136	71	4.521	2.341	0.020
Palau has the needed capacity to manage no-take and domestic fishing zones within the PNMS	100	3.140	67	4.134	4.066	< .001

Only statistically significant findings are reported. Answers of "don't know" excluded

Table 11 illustrates the results of a spearman correlation analysis conducted to determine if obtaining information regarding the PNMS from different information sources is correlated with perceptions of the PNMS. Shaded results indicate a statistically significant correlation. Positive values indicate a positive correlation, whilst negative values indicate a negative correlation. These results indicate that obtaining information from fact sheets and school outreach programs is correlated with more positive perceptions of the PNMS, whereas information provided via TV and social media are correlated with negative perceptions of the PNMS. The negative correlations are, however, not significant.

Table 11. Spearman Correlation Analysis of Information Source and PNMS Perceptions.

	Fact sheets	School programs	TV	Newspaper	Radio	Community meeting	Symposia	Internet social media
PNMS helps Palau attract quality and eco-friendly tourists.	0.142	0.181	0.104	-0.001	0.097	0.152	0.114	0.038
My household has a sustained, safe and nutritious supply of seafood.	-0.083	0.15	0.09	-0.034	0.167	-0.013	-0.018	-0.136
Offshore pelagic waters and resources are not as important as the nearshore reef and resources for the well-being of Palauans	0.240	0.083	0.128	-0.04	0.135	0.153	0.259	-0.002
Offshore pelagic fish is affordable food for my household.	-0.033	0.146	-0.155	-0.141	-0.132	0.021	-0.045	-0.022
PNMS helps conserve reefs to stay healthy for dive and tourism industries	0.212	0.149	0.083	-0.046	0.145	0.186	0.071	-0.034
PNMS helps the reefs to stay strong to better protect the coastline and coastal communities.	0.175	0.202	0.124	-0.005	0.237	0.081	0.091	-0.007
Permanent closure or no-take from fisheries by PNMS helps Palau sustain and replenish its fisheries.	0.228	0.217	-0.049	-0.082	-0.088	0.036	0.063	0.028
Currently PNMS has no benefit for my household	-0.079	-0.074	-0.150	-0.229	-0.087	-0.12	-0.116	-0.13
PNMS helps revive and strengthen our traditional connection to our environment and resources	0.212	0.172	-0.065	-0.058	0.061	0.146	0.122	-0.037
PNMS helps maintain Palau's marine resources that are important to our sense of place and identity	0.178**	0.163	-0.061	-0.036	0.035	0.083	0.099	-0.084
PNMS management has a good process to involve communities in their decision and activities	0.170	0.182	-0.048	-0.112	0.031	0.127	0.079	-0.164
PNMS makes it difficult for my household to fish the way we used to	-0.037	-0.206	0.022	0.005	-0.088	-0.138	0.004	0.021
PNMS benefits the next generations	0.162	0.184	-0.063	0.01	0.054	0.133	0.075	-0.021
PNMS creates the next generation of Palauan leadership to manage Palau's open ocean resources	0.183	0.156	-0.002	2.576e -4	0.064	0.038	0.027	-0.044
PNMS makes the conservation efforts of Palau known internationally	0.108	0.084	-0.038	-0.058	-0.016	0.060	0.072	-0.008
Palau has the needed capacity to manage no-take and domestic fishing zones within the PNMS	0.204	0.220	0.137	-0.061	0.265	0.104	0.068	-0.047

Shaded = statistically significant

Discussion

A standard approach was developed and tested to gain a better understanding of the human dimensions of the Palau National Marine Sanctuary. This included the development of a locally significant household questionnaire to provide a baseline understanding of the status of fishing and marine resource use, community perceptions regarding potential benefits from the PNMS and community knowledge and support for the PNMS.

Based on the results of the survey, there is a higher preference for reef fish compared to offshore pelagic fish. This preference seems to be due to the availability and affordability of reef fish compared to pelagic fish. According to Wabnitz et al. (2018), the high rate of reef fish consumption is one of the major drivers of future ecosystem degradation in Palau. The continued pressure on reef fish resources as well as impacts from climate change have the potential to significantly affect Palau's future tourism revenue (Wabnitz et al., 2018). Dacks et al. (2020) found that, pre-PNMS, the majority of reef fish were consumed locally by residents, the majority of tuna were exported internationally, and about half of resident reef fish consumption were non-commercial. This study found that limited availability and unaffordability of pelagic fish were main reasons driving reef fish preference.

It should, however, be noted that since the enactment of the PNMS and the onset of COVID-19 in 2020, both the tourism and local fishing industry scene have changed. For example, this survey was conducted prior to the establishment of Belau Offshore Fishing Incorporated. Since its establishment, fresh high-grade tuna is available at least twice a month at \$8/pound. This cost is still much higher than the \$2-\$2.40/pound usually asked for reef fish, but the availability of tuna and other pelagic fish have increased. To help alleviate pressure on reef resources, the government of Palau developed various regulations: RPPL No. 10-54 bans the export of all reef resources from Palau and Presidential Directive No. 19-36 support the Choose Pelagics initiative by requiring pelagic fish to be served at all government functions. Limited information is, however, available to determine any changes of pressure on reef resources post-PNMS and to untangle the impact of the PNMS, the various laws and Covid-19 on resource

use. Monitoring of reef fish harvesting and value chains are required to understand these changes. The pressure on reef fish resources has likely not yet decreased with the establishment of the PNMS, due to the short period this LSMPA has been in full effect, and the development of a viable domestic pelagic fishery that can provide consistent and affordable pelagic fish is required. Ongoing monitoring of fishery value chains and fish consumption patterns, focused on both reef and pelagic fish resources, are required to determine if a shift in pressure on reef fish resources will take place.

Eight human well-being domains were identified as being most relevant to the PNMS: sustainable livelihood, food security, economic well-being (income), equitable benefits to all, safety and security of life and property, and culture. Interestingly, the role of the PNMS in maintaining a healthy ocean and marine resources were most understood and agreed upon by respondents. Maintaining marine resources were also noted during a study focused on conceptualising social outcomes of LSMPAs, whereby Palauan fishermen indicated that they believe pelagic stocks will expand and become more available to local fishermen with the establishment of the PNMS and the Domestic Fishing Zone (Gruby et al., 2017). Respondents did, however, appear to have less understanding of the multiple other objectives of the PNMS, including the development of a domestic pelagic fishery and related benefits to Palau's economy, even though food security and economic well-being were identified as pertinent human-wellbeing domains. This indicates that education and outreach efforts should be more strategically focused on each individual objective and related outcomes. Despite the limited understanding of all potential benefits from the PNMS and how it relates to the household level, respondents showed a high level of support for the PNMS.

Those respondents that personally fished showed a greater appreciation for the conservation related benefits of the PNMS and that the 80% offshore MPA will benefit fish stocks as well as nearshore environments. In addition, the value of the PNMS in supporting future conservation leaders and benefiting the next generation were supported more by those respondents that use fish for customary and cultural purposes. Despite this understanding, some misperceptions regarding the PNMS were found. For example, some respondents felt that Palauans were being restricted from fishing in their

own waters with the establishment of the PNMS. This is, however, not the case, as offshore fishing by artisanal fishers is allowed in the Artisanal Fishing Zone. The equitable distribution of benefits amongst all Palauans were also questioned, and some respondents stated that mostly those who have the means to access the PNMS, including high government officials, will benefit the most from the PNMS. These results are a clear indicator of the lack of understanding stemming from some community members, and that effective communication will be key in ironing out these misconceptions.

The limited knowledge of the PNMS by respondents clearly indicates a need for more effective and targeted education and outreach, specifically in understanding both direct and indirect benefits of the PNMS. Future communication and outreach initiatives should focus on the following PNMS benefits: marine resource conservation, the development of Palau's offshore domestic fishing fleet, increased EEZ monitoring and surveillance, Palau still being allowed to sell its tradable vessel days through the Parties to the Nauru Agreement (PNA), domestic ownership of fishing rights to the DFZ, and increased revenue from higher tourist entry fees (Gruby et al. 2017). Results from this study also show that the most commonly used form of information gathering by respondents is through television and radio. Increasing awareness through these mediums would perhaps prove more effective in delivering key messages of the PNMS.

Conclusion

In conclusion, the overall results of this study indicate that respondents have some general idea or understanding of some of the objectives of the PNMS. As the PNMS involves various sectors and multiple objectives, strategic communication of individual objectives and potential outcomes is crucial. Some of the main findings of this study include:

- Preference for pelagic fish consumption may potentially increase with an increase in consistent availability of affordable pelagic fish.
- There is a strong need for outreach/awareness and education to all levels of the community focused on addressing misperceptions.

The development and testing of a standard PNMS household survey provides an approach to monitor the human dimension of the PNMS. Specifically, future changes to fishing and fish use patterns can be compared to findings from this study, changes of perceptions of PNMS benefits can be monitored and the effectiveness of future education and outreach initiatives can be tracked. The next phase of this research will include additional PNMS household surveys within the entire Palau, to provide a robust baseline understanding of some of the human dimensions of the PNMS for the people of Palau.

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Annex A

Table A1: Total percentage scores of potential PNMS benefits as perceived by respondents (n=266). SD = Strongly Disagree; D = Disagree; N = Neither; A = Agree; SA = Strongly Agree; DK = Don't Know, N/A = Not Applicable.

Statement	SD	D	N	A	SA	DK	N/A
PNMS helps Palau attract quality and eco-friendly tourists.	5.6	3.4	4.9	25.6	33.5	23.3	3.8
My household has a sustained, safe and nutritious supply of seafood.	2.3	6.0	0.8	13.2	56.8	16.2	4.9
Offshore pelagic waters and resources are not as important as the nearshore reef and resources for the well-being of Palauans.	36.5	9.4	4.1	9.4	17.3	19.9	3.4
Offshore pelagic fish is affordable food for my household.	36.8	9.8	1.9	13.5	10.9	18.8	8.3

PNMS helps conserve reefs to stay healthy for dive and tourism industries	5.6	6.0	1.5	25.9	32.0	25.6	3.4
PNMS helps the reefs to stay strong to better protect the coastline and coastal communities.	5.6	4.1	2.3	22.2	36.8	25.6	3.4
Permanent closure or no-take from fisheries by PNMS helps Palau sustain and replenish its fisheries.	10.9	4.1	4.9	17.7	35.3	23.3	3.8
Currently PNMS has no benefit for my household	16.2	9.0	3.8	16.9	22.6	28.2	3.4
PNMS helps revive and strengthen our traditional connection to our environment and resources	6.4	3.8	4.1	20.7	32.3	28.9	3.8
PNMS helps maintain Palau's marine resources that are important to our sense of place and identity	5.3	3.4	3.0	24.8	35.3	24.8	3.4
PNMS management has a good process to involve communities in their decision and activities	12.8	7.9	3.8	13.9	26.3	30.5	4.9
PNMS makes it difficult for my household to fish the way we used to	27.1	10.2	3.4	6.4	4.9	32.3	15.8
PNMS benefits the next generations	5.3	3.0	1.5	21.8	42.9	22.2	3.4
PNMS creates the next generation of Palauan leadership to manage Palau's open ocean resources	5.3	2.6	3.8	22.6	38.7	23.3	3.8
PNMS makes the conservation efforts of Palau known internationally	4.5	1.1	3.0	21.1	38.3	28.6	3.4
Palau has the needed capacity to manage no-take and domestic fishing zones within the PNMS	13.5	6.4	2.6	13.2	27.1	32.7	4.5